



Project Proposal

CSC-225– Software Engineering

Department of Computer Sciences

Namal University

Mianwali, Pakistan

Project Title: Namal Student Assistance Provider (NSAP)

V 1.0

BY:

1	<i>Laiba Hussain</i>	<i>NUM-BSCS-2025-23</i>
2	<i>Alishba Sana</i>	<i>NUM-BSCS-2025-04</i>
3	<i>Madiha Akram</i>	<i>NUM-BSCS-2025-24</i>
4	<i>Sehrish</i>	<i>NUM-BSCS-2025-52</i>

Submission Date: 9th October, 2026

REVISION HISTORY

Date	Version	Description	Approved by
9 October 2026	V 1.0	Initial version of the NSAP project proposal submitted for approval.	

TABLE OF CONTENTS

1. INTRODUCTION:	5
2. PROBLEM STATEMENT:.....	5
3. PROJECT OBJECTIVES:	5
4. PROJECT SCOPE:	6
5. STAKEHOLDER IDENTIFICATION:	6
6. SOFTWARE DEVELOPMENT METHODOLOGY:	6
7. TOOLS AND TECHNOLOGIES:.....	9
REFERENCES	9

REQUIREMENT PROVIDER AGREEMENT

This agreement is made between the Requirement Provider and the Project Team for the development of the proposed software project titled:

Project Title: Namal Student Assistance Provider (NSAP)

The Requirement Provider will provide the necessary project information, domain knowledge, user needs, and other relevant details required for understanding the proposed system and will review and validate the information documented by the Project Team. The Project Team will accurately document the agreed project scope and develop the system based on the requirements established and approved during the subsequent requirements engineering process. Any significant changes to the agreed project scope will be discussed with the Requirement Provider and Project Team before implementation. Both parties agree to maintain effective communication and cooperation throughout the project.

Requirement Provider

Name: Mr. Haris Bilal

Designation: Assistant Registrar

Organization: Namal University Mainwali

Email: asst.registrar@namal.edu.pk

Project Team Representative

Name: Laiba Hussain

Roll No: NUM-BSCS-2025-23

Email: bcs25f23@namal.edu.pk

Signature with Date:


8/10/2026

Signature with Date:


8/10/2026

1. INTRODUCTION:

NSAP is a web-based record management system that we are developing for the Student Support Office (SSO) of the university. At present, the SSO keeps student records in files and physical documents, which takes a lot of time and is difficult to keep organized. NSAP will move this work to one online platform where students can submit their own information and the SSO staff can review and manage it.

The system has two sides. On the student side, every student logs in to a personal account, fills in the required fields and submits the record to the SSO. On the SSO side, the staff reviews the submitted records. If a document is missing, the individual is informed directly through the same platform. A student cannot change approved data on their own and if student want to change something then he/she will submit a request to SSO and SSO can approve or reject it. The data of a student can be seen only by that student and by the SSO staff, and nobody else can access it. The SSO can also post important announcements for the students. To find a record, the SSO staff selects the year, then the department, and then the student from the list arranged by registration number.

2. PROBLEM STATEMENT:

The SSO has to manage the records of a large number of students from different departments and academic years. When this data is kept in separate files and physical documents, finding one student's record, checking that all documents are complete and updating information take a lot of time and effort. Students also have to submit their documents physically and may not know that something is missing until the office contacts them. In addition, no proper process for correcting data that has already been approved. A digital system with controlled access can solve these problems for both the students and the SSO staff.

3. PROJECT OBJECTIVES:

The main objectives of the proposed system are:

- To develop a web-based system for managing the student records of the SSO digitally.
- To give every student a platform where they can enter their personal and academic information in the required fields and submit it for review.
- To allow the SSO staff to review submitted records and inform students about missing documents directly through the system.
- To keep student data private, so that it can be viewed only by the student and the SSO staff.
- To organize records year-wise and department-wise, with students listed by registration number, so that any record can be found quickly.
- To make students send a change request for any change in approved data, and to allow the change only after the SSO accepts the request.
- To allow the SSO to post important announcements for students.
- To reduce manual paperwork and save the time of both students and SSO staff.

4. PROJECT SCOPE:

The proposed system will manage student records digitally and will provide a direct way of communication between the students and the SSO. Its main modules are:

- **Student Management:** login, personal profile, data entry and submission of documents.
- **Department and Year-wise Records:** records arranged by year, then department, then registration number.
- **SSO Management:** review of records, missing document alerts, and acceptance or rejection of change requests.
- **Notification Management:** announcements and notifications from the SSO to students.

In this semester, the team will design only the front end (user interface) of the system. The back end and the database will be developed in the next semester. The system will not cover complete academic management or fee management. Communication on the platform will be only between the SSO and the concerned student

5. STAKEHOLDER IDENTIFICATION:

The major stakeholders of the proposed system are the students, the SSO staff and the project team.

Table 1: Stakeholders and Their Roles in the Proposed System

Stakeholder	Role	Interaction with System
Students	End User	Login Enter and submit personal and academic information View own profile and record status Send change requests Receive missing document alerts and announcements
SSO Staff	System Administrator	Review submitted records View records year-wise and department-wise Inform students about missing documents Accept or reject change requests Post announcements
Project team	Developers	Design Develop Test Maintain and update

6. SOFTWARE DEVELOPMENT METHODOLOGY:

The proposed system will follow the Agile software development methodology, and the Scrum framework will be used to manage the work in short sprints. Agile is suitable for this project because the needs of the SSO and the students may change during development. For example, new notification types, profile fields or record management features may be requested after the first version is shown to them. Working in small increments allows the team to take feedback after every sprint, find problems early and improve the system step by step.

The development will start with requirements gathering and analysis to identify the required screens and features. In the design phase, the team will prepare blueprint and the navigation

structure of the system. The main front-end modules are student login, student profile, record status, change request, notifications and announcements, and the SSO dashboard with year-wise and department-wise student records. After implementation, the screens will be tested for functionality, usability, responsiveness and navigation. Feedback from the instructor and the team members will be used to improve the front end before the final submission of this semester. In the next semester, the team will develop the back end and the database and connect them with the approved front end.

6.1 Scrum-Based One-Year Development Plan

The project will be completed in twelve months. The first six months are for the user interface design and the next six months are for the database and complete system development. The sprint plans are given in Tables below;

First Six Months: UI Development

Table 2: Sprint Plan for the First Six Months (UI Development)

Phase / Sprint	Scrum-Aligned Description
Product Backlog Creation	Initial Phase: Meeting with the Requirement Provider (RP) to identify the needs of students and SSO staff. User Stories will be created for the main system features.
Sprint 0: UI Design	Initial Setup: Creating wireframes and designing the main screens for students and SSO staff. The navigation and overall layout will be planned.
Sprints 1–2: Student Interface	Student Features: Designing the student login, profile, data entry form, document submission and change request screens.
Sprints 3–4: Staff Interface	SSO Features: Designing screens for the SSO staff to view student records year-wise and department-wise, review submissions, mark missing documents, and accept or reject change requests.
Sprints 5–6: Communication Interface	Communication: Designing the announcement, notification and missing document alert screens. The complete UI will be reviewed and improved using RP feedback.

Second Six Months: Database and Complete System Development

Table 3: Sprint Plan for the Second Six Months (Database and System Development)

Phase / Sprint	Scrum-Aligned Description
Sprints 7–8: Database Development	Database Setup: Creating the database structure and adding the required student, staff, document, change request and announcement data.

Phase / Sprint	Scrum-Aligned Description
Sprints 9–10: System Integration	Integration: Connecting the user interface with the database so that student and staff information can be stored, viewed, and updated.
Sprints 11–12: Data & System Features	Complete Features: Adding and managing the required data, connecting the forms with the system, and making sure that change requests, missing document alerts, announcements and other features work correctly.
Sprints 13–14: Testing & Refinement	Testing: Testing the complete system, checking data handling, user access, navigation, forms, and fixing errors found during testing.
Sprints 15–16: Finalization & Deployment	Final Phase: Conducting User Acceptance Testing (UAT), making final improvements, completing documentation, and preparing the complete NSAP website for final deployment and demonstration.

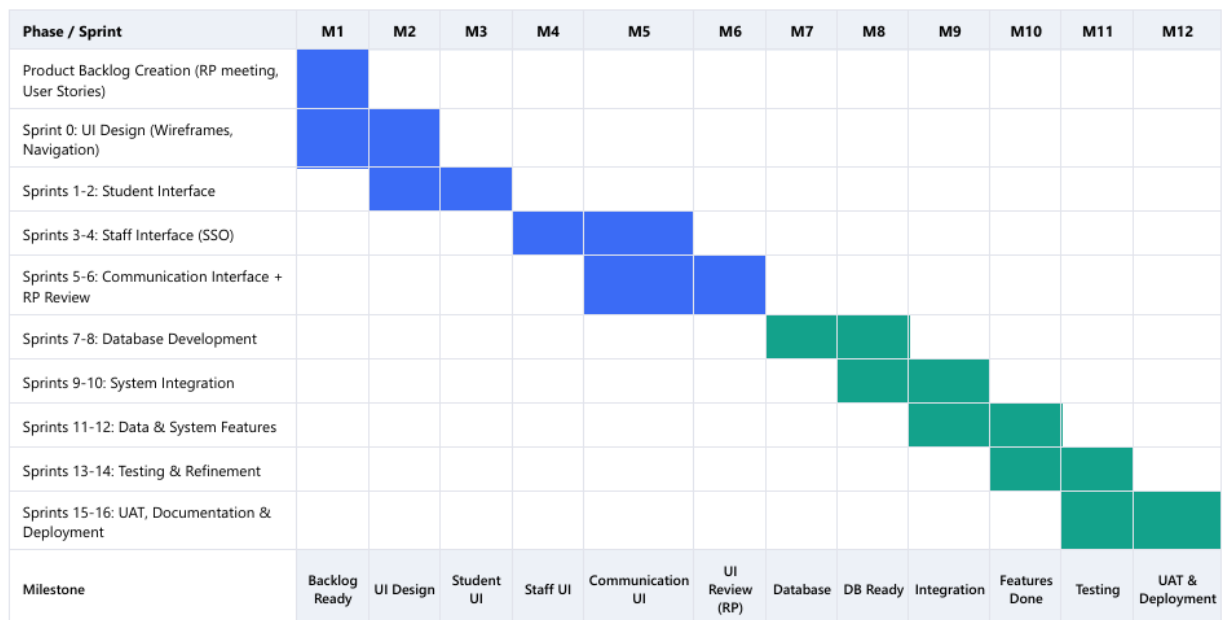


Figure 1: Project Timeline of NSAP (Gantt Chart)

7. TOOLS AND TECHNOLOGIES:

Table below lists the tools and technologies that we use for our project.

Table 4: Proposed Tools and Technologies

#	Tools and Technology	Purpose
1	Figma	Designing wireframes and clickable prototypes of the student and SSO screens.
2	HTML5	Building the structure of the web pages.
3	CSS3	Styling the pages and making the layout responsive.
4	JavaScript	Adding interaction and form validation to the user interface.
5	Visual Studio Code	Writing and editing the front-end code.
6	Git and GitHub	Version control and sharing the code and other project related data between team members.
7	Node.js with Express.js	Will be used to develop the back end
8	MySQL	Storing student, staff, document, request and announcement data in the next semester.

REFERENCES

- [1] OpenAI, “Chat Gpt,” [Online]. Available: <https://chatgpt.com/share/6ac92f8c-4db8-83ee-bc0c-95392750a877>. [Accessed: Oct. 7, 2026].
- [2] Anthropic, “Claude,” [Online]. Available: <https://claude.ai/share/7d918ce4-b241-4df0-82aa-796488ed69b6>. [Accessed: Oct. 8, 2026].
- [3] K. Schwaber and J. Sutherland, “The Scrum Guide,” Nov. 2020. [Online]. Available: <https://scrumguides.org/scrum-guide.html>. [Accessed: Oct. 9, 2026].
- [4] Bareera Ijaz, *Student Project Proposal (Namal Criko)*, BS Computer Science, Namal University, 2024.